



CS 540 Introduction to Artificial Intelligence
Linear Algebra and PCA
University of Wisconsin-Madison

Fall 2026 Sections 1 & 2

Announcements

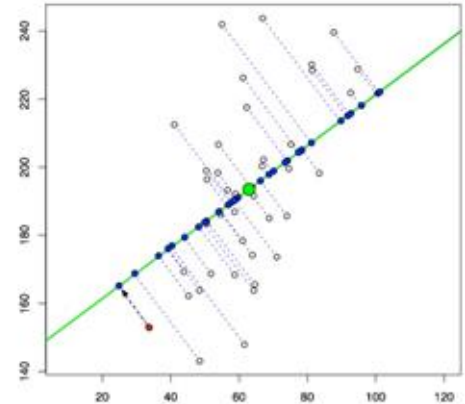
- **HW 1 released:**
 - Due **Friday Sep 18 at 11:59PM**

- Class roadmap:

Linear Algebra & PCA
NLP
Machine Learning Introduction

Outline

- Basics: vectors, matrices, operations
- Dimensionality reduction
- Principal Components Analysis (PCA)



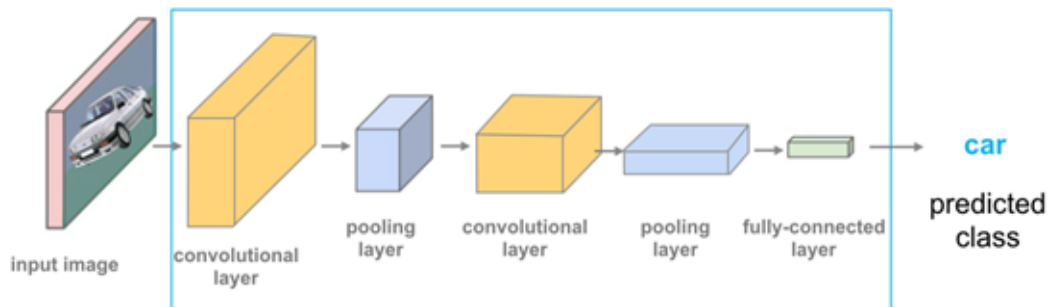
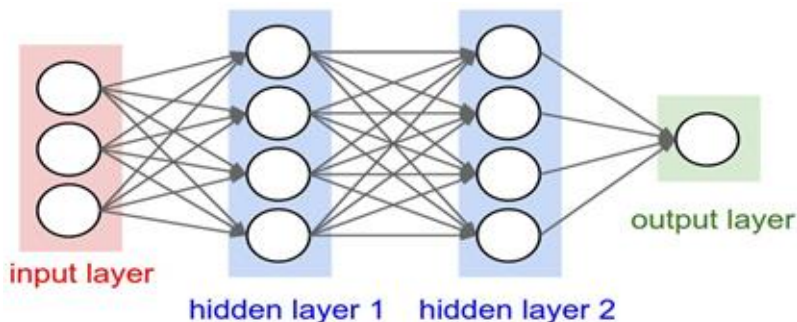
Lior Pachter



Linear Algebra

Review: Linear Algebra

- In AI/ML: building blocks for **all models**
 - e.g., linear regression; part of neural networks
- Vectors: list of values; point in space
- Matrices: Table of values; list of vectors



Matrix & Vector Operations

- **Matrix-Vector multiplication:**
 - Linear transformation; plug in vector, get another vector
 - Each entry in Ax is the inner product of a row of A with x

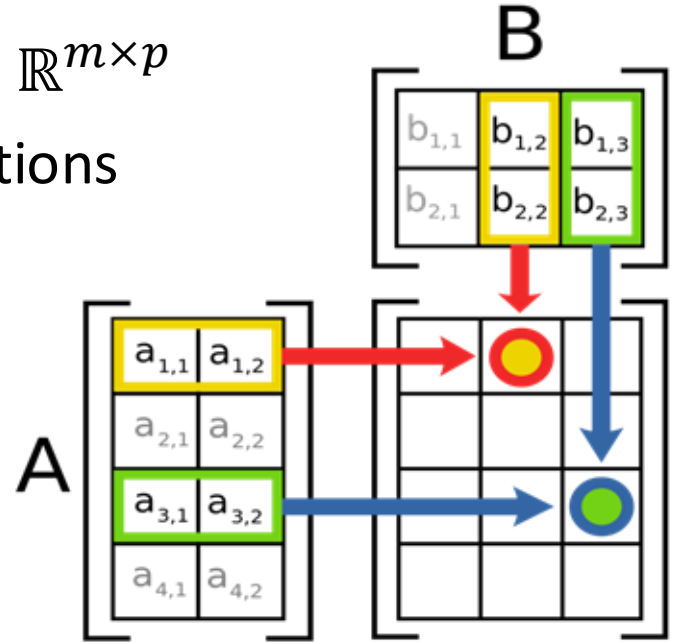
$$x \in \mathbb{R}^n, A \in \mathbb{R}^{m \times n}$$

$$Ax = \begin{bmatrix} \langle A_{1:}, x \rangle \\ \langle A_{2:}, x \rangle \\ \vdots \\ \langle A_{m:}, x \rangle \end{bmatrix} = \begin{bmatrix} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n \\ \vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n \end{bmatrix}$$

Matrix & Vector Operations

- Matrix multiplication
 - $A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times p}$, then $AB \in \mathbb{R}^{m \times p}$
 - “Composition” of linear transformations
 - Not commutative in general!

$$AB \neq BA$$



Identity Matrix

- Like “1”
- Multiplying by it gets back the same matrix or vector
- Rows & columns are the “**standard basis vectors**” e_i

$$I = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

$\downarrow \quad \downarrow \quad \quad \downarrow$
 $e_1 \quad e_2 \quad \quad e_n$

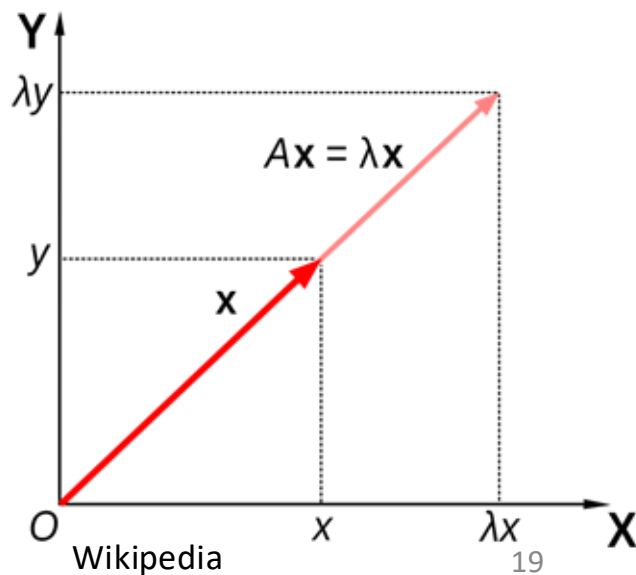
Matrix Inverses

- If for A there is a B such that $AB = BA = I$
 - Then A is invertible/nonsingular, B is its inverse
 - Some matrices are **not** invertible!
 - Usual notation: A^{-1}

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \times \begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix} = I$$

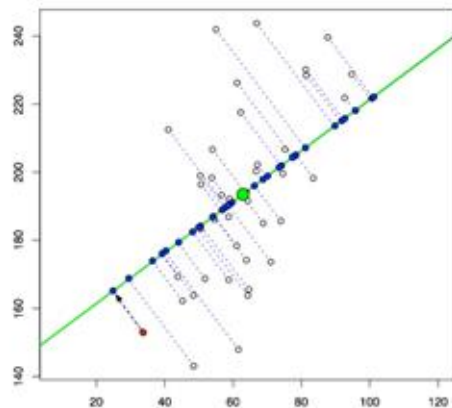
Eigenvalues & Eigenvectors

- For a square matrix A , solutions to $Av = \lambda v$
 - v (nonzero) is a vector: **eigenvector**
 - λ is a scalar: **eigenvalue**
 - Intuition: A is a linear transformation;
 - Can stretch/rotate vectors;
 - Eigenvectors: only stretched by eigenvalues)



Outline

- Basics: vectors, matrices, operations
- **Dimensionality reduction**
- Principal Components Analysis (PCA)

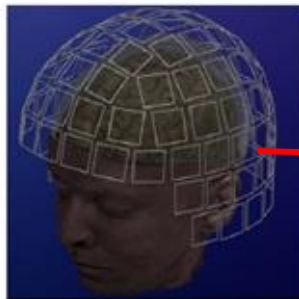


Lior Pachter

Dimensionality Reduction

- Vectors store features. Lots of features!
 - Document classification: thousands of words per doc
 - Netflix surveys: 480189 users x 17770 movies
 - **MEG Brain Imaging**: 120 locations x 500 time points x 20 objects

	movie 1	movie 2	movie 3
Tom	5	?	?
George	?	?	3
Susan	4	3	1
Beth	4	3	?



Dimensionality Reduction

Reduce dimensions

- Why?
 - Lots of features redundant
 - Storage & computation costs
- Goal: take $x \in \mathbb{R}^d \rightarrow x \in \mathbb{R}^r$ for $r \ll d$
 - But minimize information loss

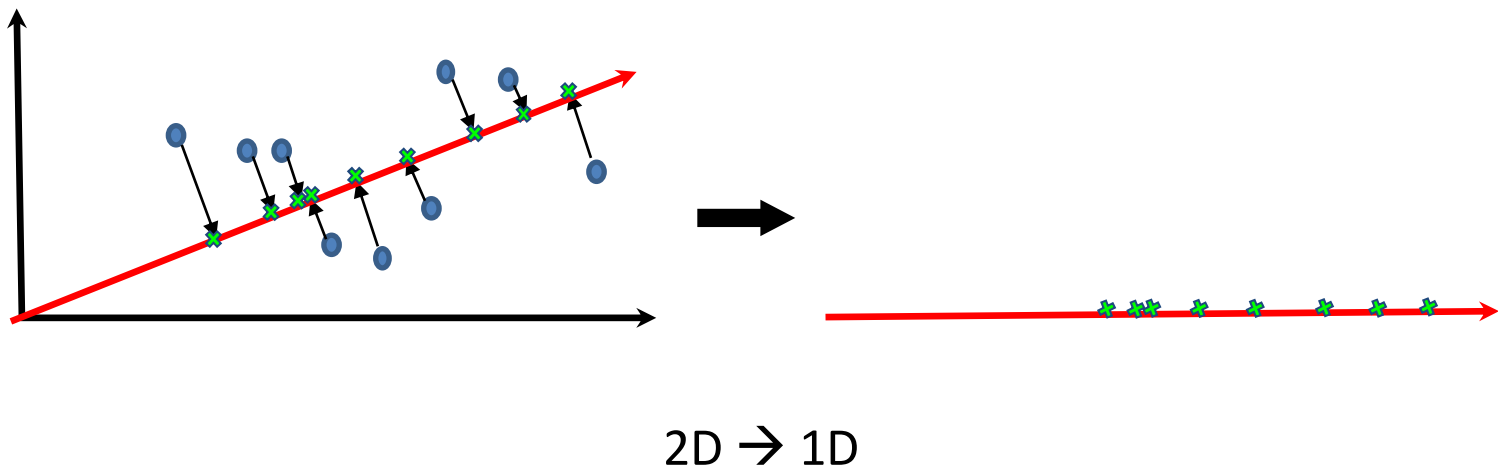


CreativeBloq

Principal Components Analysis (PCA)

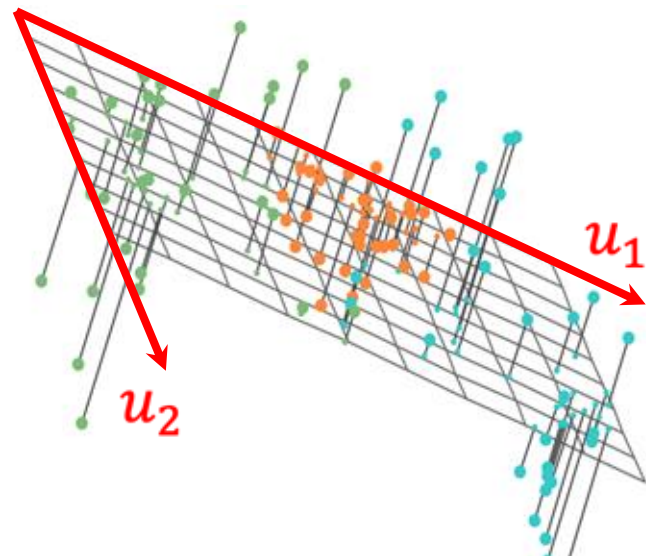
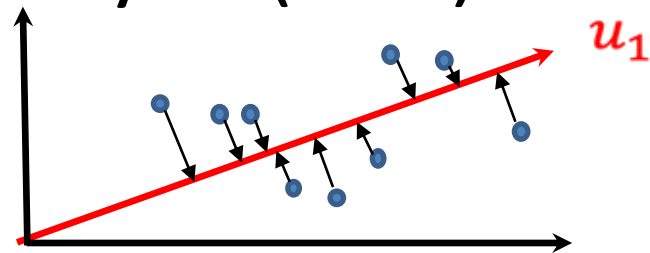
A type of dimensionality reduction approach

→ For when data is **approximately lower dimensional**



Principal Components Analysis (PCA)

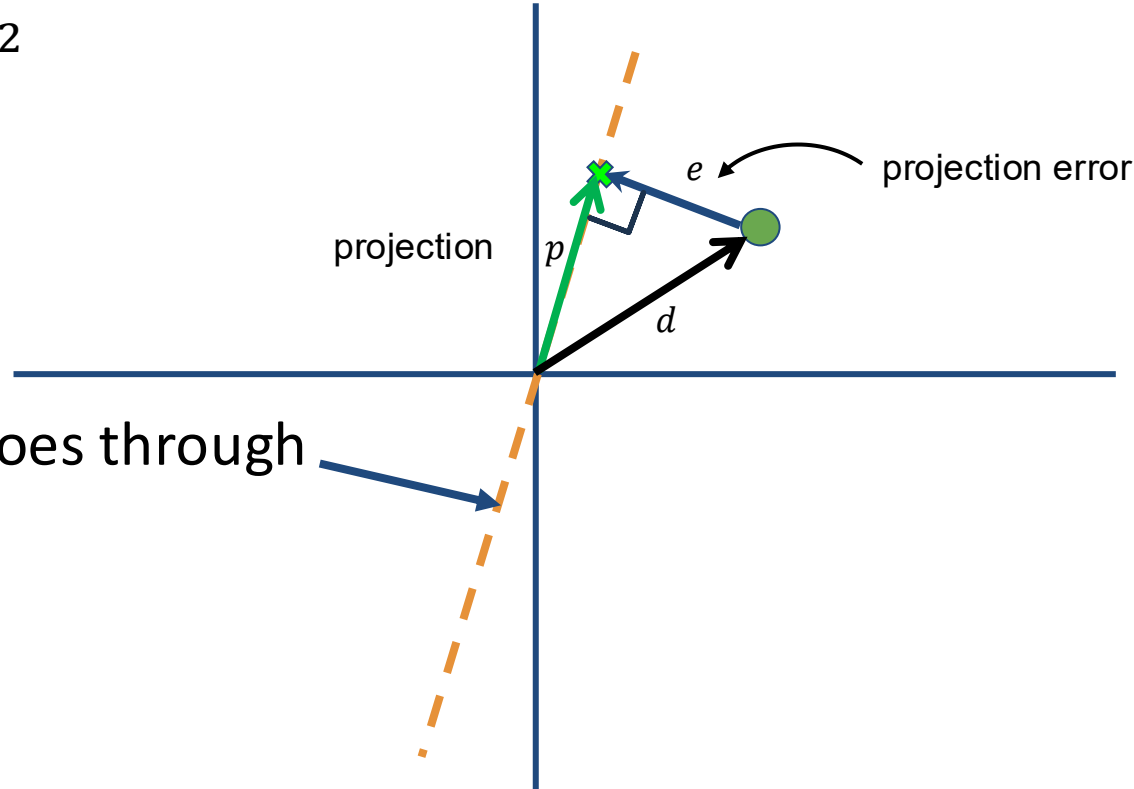
- Find axes $u_1, u_2, \dots, u_n \in \mathbb{R}^d$ of a subspace
 - Will project to this subspace
- These vectors are the **principal components**
- Want to preserve data
 - Minimize projection error



Projection: An Example

$$x \in \mathbb{R}^2$$

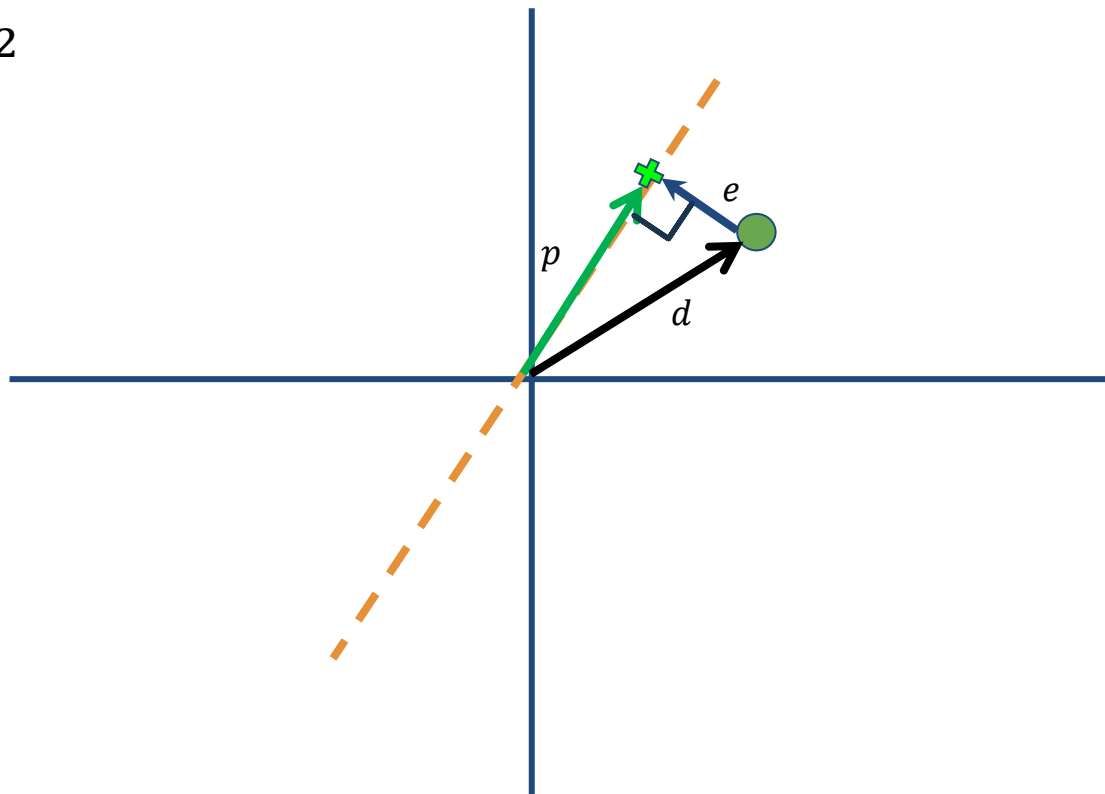
$$d^2 = p^2 + e^2$$



A line that goes through
the origin

Projection: An Example

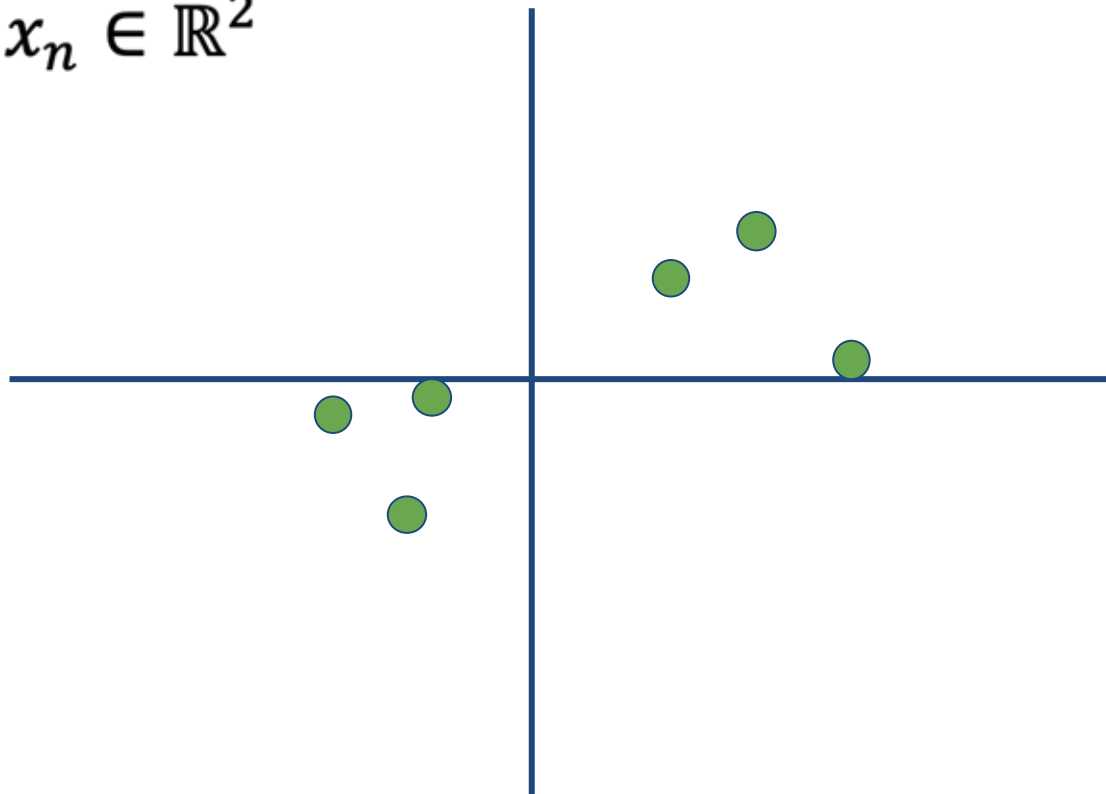
$x \in \mathbb{R}^2$



$$d^2 = p^2 + e^2$$

Projection: An Example

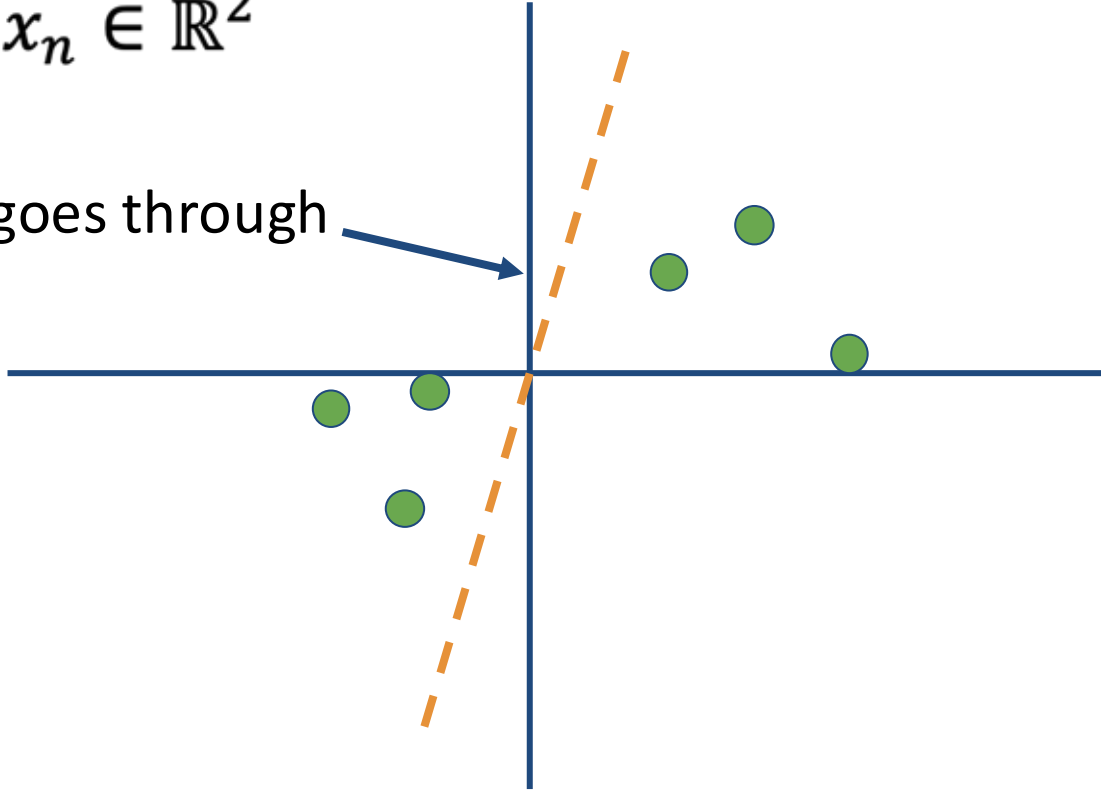
$$x_1, x_2, \dots, x_n \in \mathbb{R}^2$$



Projection: An Example

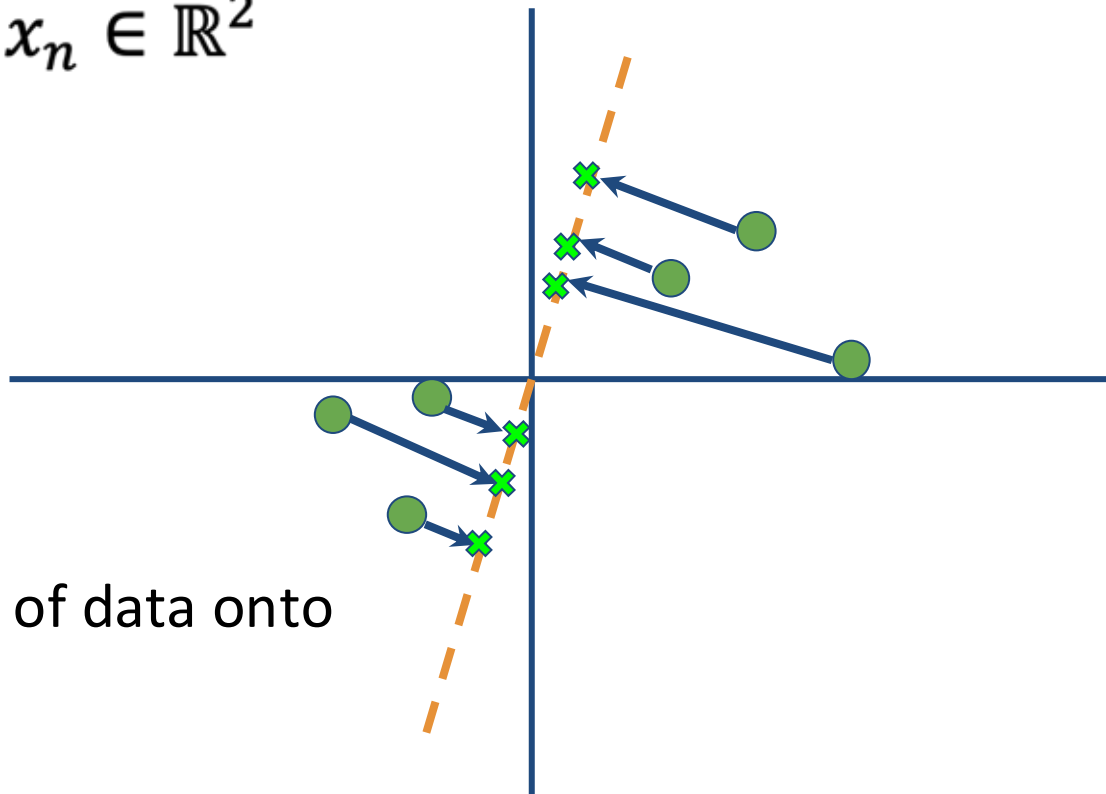
$$x_1, x_2, \dots, x_n \in \mathbb{R}^2$$

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Projection: An Example

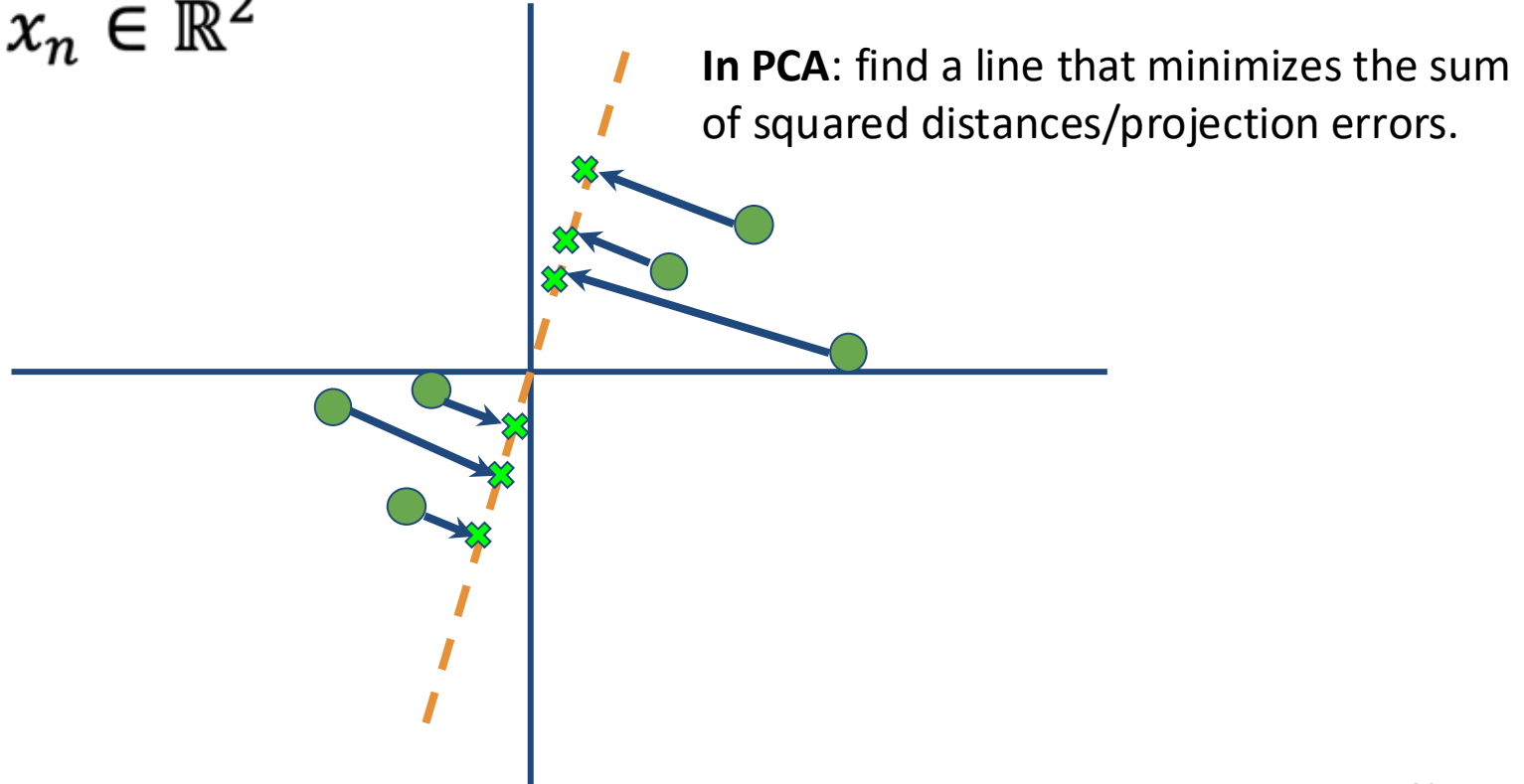
$$x_1, x_2, \dots, x_n \in \mathbb{R}^2$$



Projection of data onto
this line

Projection: An Example

$$x_1, x_2, \dots, x_n \in \mathbb{R}^2$$

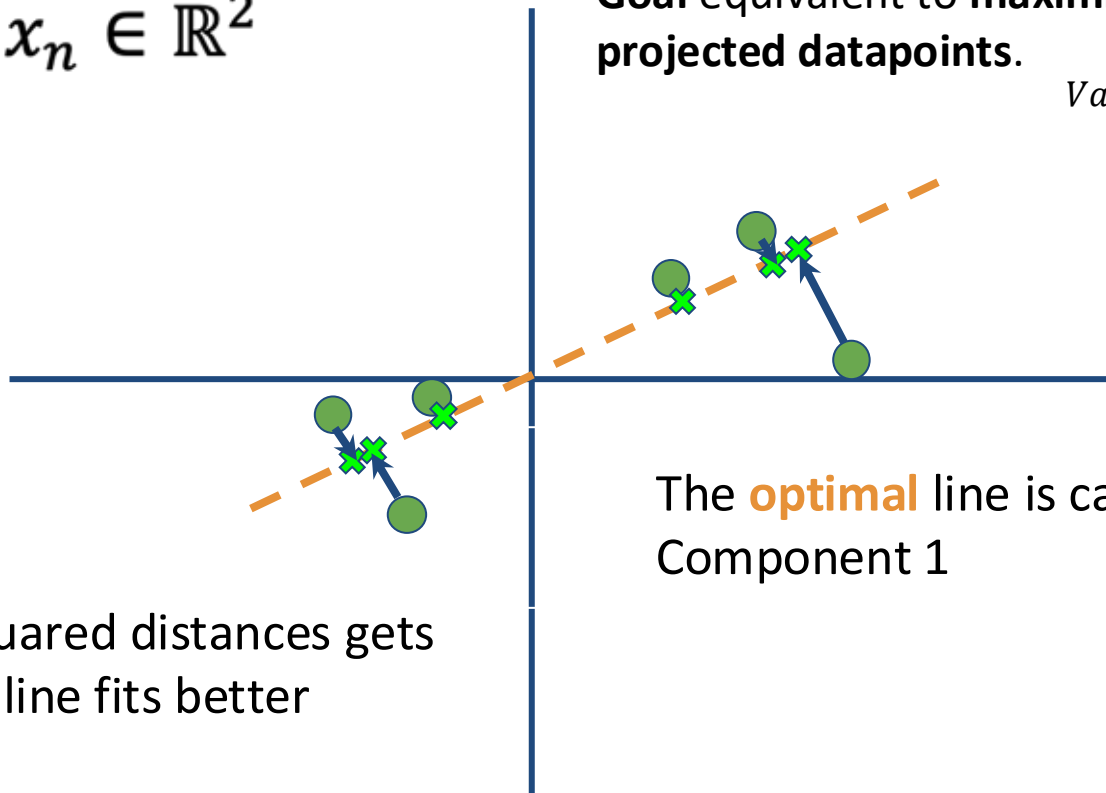


Projection: An Example

$$x_1, x_2, \dots, x_n \in \mathbb{R}^2$$

Goal equivalent to **maximizing the variances of projected datapoints.**

$$Variance = \frac{1}{n-1} \sum_{i=1}^n \langle u, x_i \rangle^2$$



The sum of squared distances gets smaller as the line fits better

The **optimal** line is called Principal Component 1

PCA Procedure

Input: data $x_1, x_2, \dots, x_n \in \mathbb{R}^d$

Step 1: Center data at the origin so that $\hat{\mu} = 0$

Step 1.1: Calculate the mean $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$

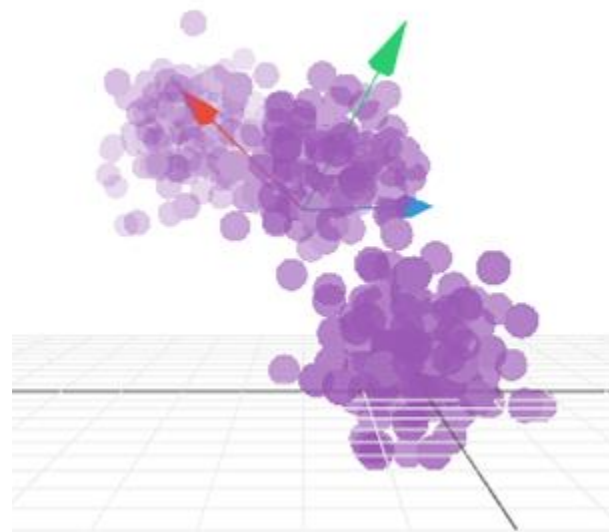
Step 1.2: Subtract the mean from the datapoints

$$x_1 = x_1 - \hat{\mu}$$

$$x_2 = x_2 - \hat{\mu}$$

....

$$x_n = x_n - \hat{\mu}$$



Victor Powell

PCA Procedure

Objective: $u = \arg \max_{\|u\|=1} \frac{1}{n-1} \sum_{i=1}^n \langle u, x_i \rangle^2$

$$\frac{1}{n-1} \sum_{i=1}^n u^T (x_i x_i^T) u$$

$$u^T \left(\underbrace{\frac{1}{n-1} \sum_{i=1}^n x_i x_i^T}_{\text{Covariance Matrix}} \right) u$$

Covariance Matrix

PCA Procedure

Output: principal components $u_1, u_2, \dots, u_m \in \mathbb{R}^d$

The **m principal components** are:

- orthogonal
- the **top- m eigenvectors** of the covariance matrix ($S = \frac{1}{n-1} \sum_{i=1}^n x_i x_i^T$)

Step 2: Compute **covariance matrix**

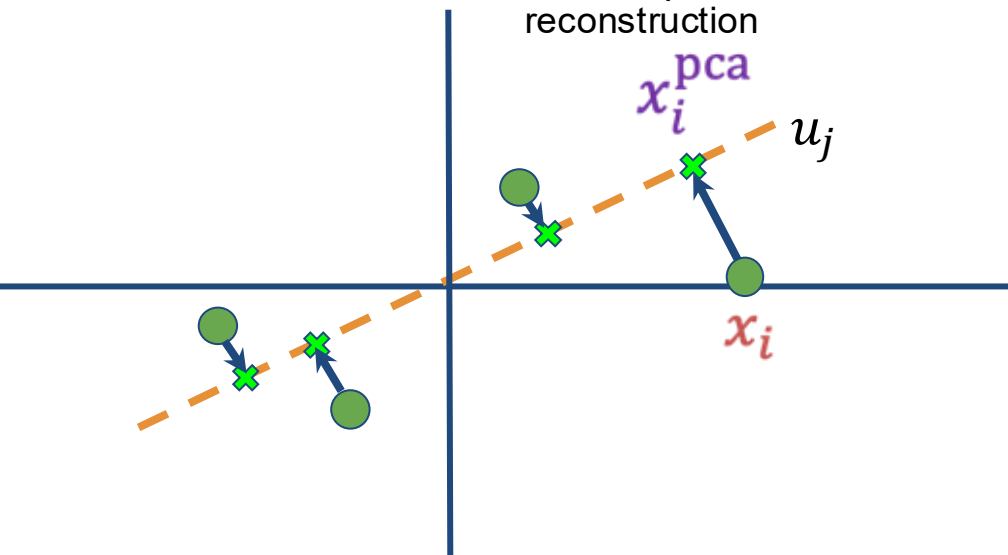
Step 3: Compute **eigenvectors** and **eigenvalues** of covariance matrix

Step 4: **Sort eigenvalues** from high to low

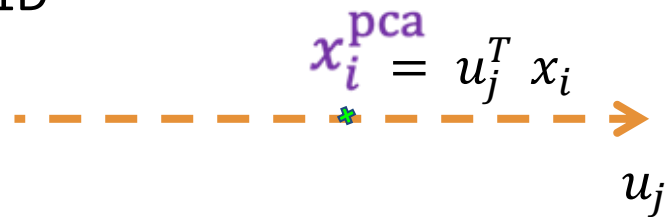
Step 5: Select **top- m eigenvectors**

PCA Projection and Reconstruction

For each x_i : $x_i^{pca} = \underbrace{\sum_{j=1}^m \overbrace{(u_j^T x_i)}^{\text{projection}} u_j}_{\text{reconstruction}}$

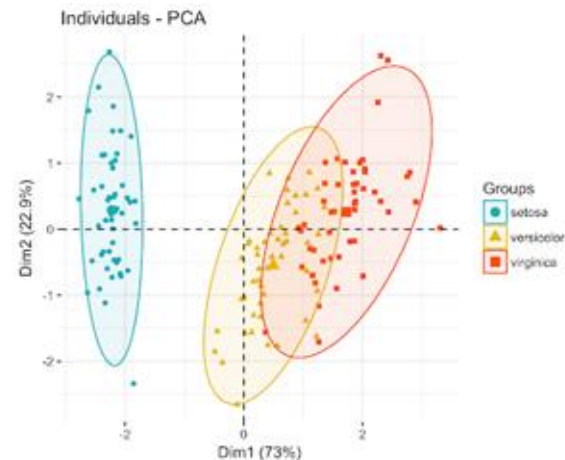


2D $\rightarrow$ 1D



Many Variations

- PCA, Kernel PCA, ICA, CCA
 - Extract structure from high dimensional dataset
- Uses:
 - Visualization
 - Efficiency
 - Noise removal
 - Downstream machine learning use



STHDA

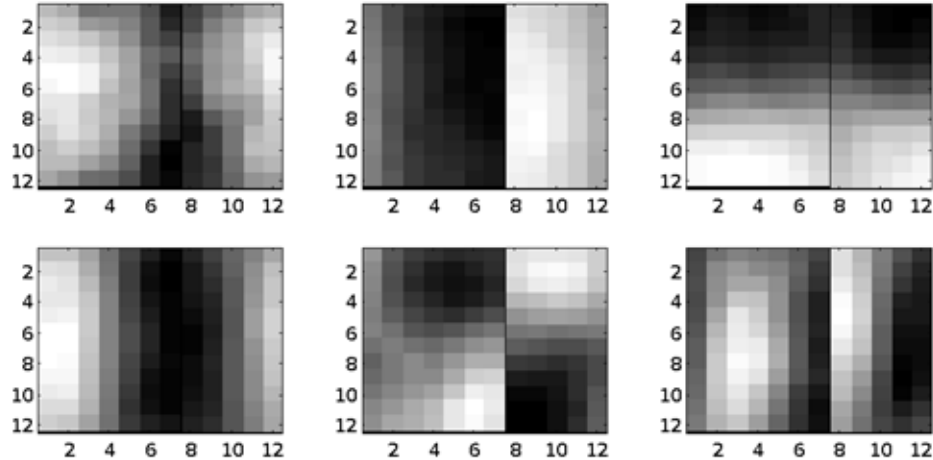
Application: Image Compression

- Start with image; divide into 12x12 patches
 - That is, 144-D vector
 - **Original image:**



Application: Image Compression

- 6 principal components (as an image)



Application: Image Compression

- Project to 6D



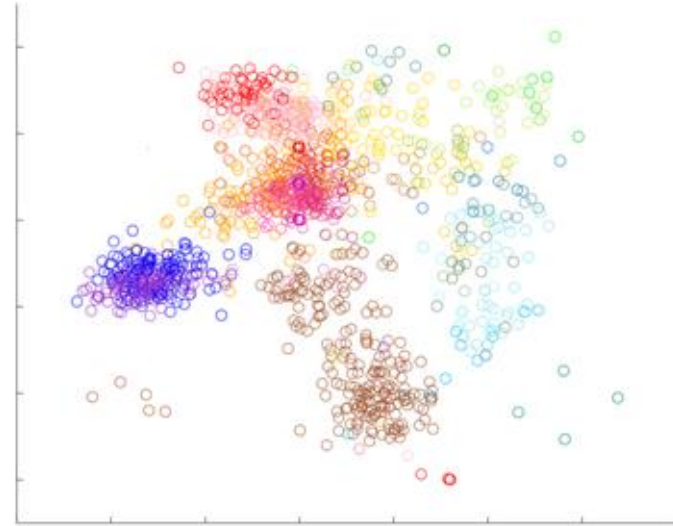
Original



Compressed

Application: Exploratory Data Analysis

- [Novembre et al. '08]: Take top two singular vectors of people x SNP matrix (POPRES)



“Genes Mirror Geography in Europe”

Readings

- Vast literature on linear algebra.
- Local class: **Math 341**
- More on PCA (and other matrix methods in ML): **CS 532**
- **Suggested reading:**
 - Textbook: Artificial Intelligence: A Modern Approach (4th edition). Stuart Russell and Peter Norvig. Pearson, 2020. Appendix A
 - Lecture notes on PCA by Roughgarden and Valiant
<https://web.stanford.edu/class/cs168/l/l7.pdf>